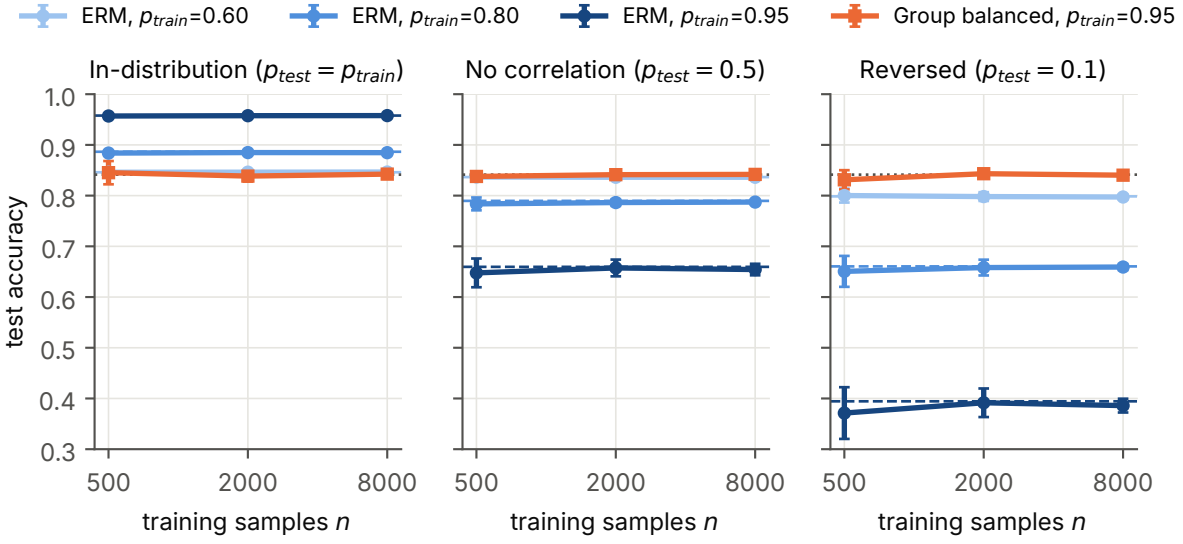
ERM, p_{train}=0.60
ERM, p_{train}=0.80
ERM, p_{train}=0.95
Group balanced, p_{train}=0.95
In-distribution ($p_{test} = p_{train}$)
No correlation ($p_{test} = 0.5$)
Reversed ($p_{test} = 0.1$)
test accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
500
2000
8000
training samples n
dashed: large-sample theory for ERM; dotted: $\Phi(1) = 0.841$ (stable-only optimum)